

Kenar Narayaka

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EDUCATION

University of Oxford, Christ Church

Oxford, England

BA in Computer Science & Philosophy – Pass in 1st-Year Exams

2025 – 2028

Relevant Modules: Probability, Linear Algebra, Discrete Maths, Design & Analysis of Algorithms, Algorithms & Data Structures, Functional Programming (Haskell), Imperative Programming (Scala)

Loughborough College

Loughborough, England

A-Level Maths, Further Maths, Computer Science – A*AA

2023 – 2025

EXPERIENCE

Hitachi Astemo

Jakarta, Indonesia

Backend Engineer Intern – Human Resources | *Fat-Free Framework, PHP, Postgres*

Jun. 2024 - Aug. 2024

- Built an HR platform for ~15,000 staff, digitising leave, overtime and manpower requests with normalised PostgreSQL schemas and dashboards
- During HR pilots, cut request turnaround 97%, from seven days to five hours, using online forms and CRUD approval dashboards
- During HR pilots, built PDF-to-HTML tool that processed ~20 form submissions weekly and routed to network printers for redundancy

PROJECTS

3D Fractal Simulator | *JavaScript, three.js, GLSL*

- Built a real-time browser renderer for fractal families using signed-distance estimation and raymarching
- Implemented distance-adaptive step sizing to cut redundant samples, raising frame rate 40% and 20% more via cone-marching
- Profiled throughput against geometric detail across camera and fractal settings; scored 69/70, top OCR NEA in school history

Basketball Jumpshot Analyser | *Rust, ONNX Runtime, FFmpeg*

- Built a local video-analysis prototype running top-down RTMPose inference through ONNX Runtime over variable-frame-rate footage
- Computed six confidence-aware biomechanics and timing metrics, achieving mean 5° angular error and 2ms mean timing error against footage
- Implemented PTS-aware multi-shot segmentation, achieving 96% precision and 92% recall across 20 manually labelled shots

Reverse Tunnelling Reverse Proxy for Minecraft Beta | *Rust, Java, Tokio, Netty, QUIC*

- Built an in-game tool that turns single-player worlds into temporary dedicated servers across 3 modes: LAN, router mapping, or QUIC relay
- Built a Rust/Java reverse-tunnelling system supporting 8 concurrent guests and 60-second session recovery; tested on 800 byte-exact streams
- Protected worlds with 5 rotating atomic backups, SHA-256-verified server downloads, and crash-safe process supervision and recovery

Obsidian Ink LaTeX Transcriber | *TypeScript, Python, PyTorch, Obsidian API*

- Built an Obsidian plugin to convert handwritten maths into editable Markdown and LaTeX through a vision API and review interface
- Protected manual edits with stable IDs and content hashes; passed 10,000 generated integrity cases across 8 edit and conflict scenarios
- Implemented local TexTeller inference and benchmarked 200 public handwritten expressions; measured 275ms median latency on 100 tests

EXTRACURRICULAR ACTIVITIES

Christ Church Computer Science Society

Jun. 2026 – Present

Treasurer

- Organised 8 events including a 40-person hackathon and termly formals, planning and costing each within a £1,000 budget

Oxford Indonesia Society

Jun. 2026 – Present

President

- Led committee planning and delivery of termly dinners and cultural events, coordinating venues, budgets, publicity, and logistics
- Revived social media and member outreach, rebuilding membership across the Indonesian student community to a consistent 25 members

CapitOx

Jan. 2026 - Apr. 2026

Quant Academy – Oxford-Man Institute

- Built a Monte Carlo pricing pipeline for a European put under a 2-component generalised-normal mixture model (Python, NumPy)
- Implemented 3 estimators with tuned proposals, cutting variance 5x and runtime 60x vs. random-walk Metropolis-Hastings (Matplotlib)

TECHNICAL SKILLS

Languages: Haskell, Scala, Python, C++, PHP, JavaScript, SQL, GLSL, Rust

Frameworks & Libraries: NumPy, Matplotlib, three.js, Fat-Free Framework, Tauri v2

Tools: Postgres, Git, Linux, tmux, Docker